COMP90050 ADBS

Week 11

Q1 RAID and Mean Time to Failure

1. What is the Mean time to failure values of different RAID systems?

- a. RAID 0 with 2 disks
- b. RAID 2 with 2 disks
- c. RAID 1 with 2 disks
- d. RAID 1 with 3 disks
- e. RAID 3 with 3 disks
- f. RAID 4 with 3 disks
- g. RAID 5 with 3 disks
- h. RAID 6 with 5 disks

MTTF = the time elapsing before a failure is experienced

- Let p = P(event A occurs), then mean time to event A = $\frac{1}{p}$

Assuming events A and B are independent:

- P(both A and B occur) = P(A) * P(B)
- P(A or B occur) ≈ P(A) + P(B)

Let p = P(one disk failure), and
MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

E.g.
for a system with 2 disks A and B, where it fails iff both disks fails:
P(system fails) = P(A and B fails) = p^2, mean time to failure = $\frac{1}{p^2} = MTTF^2$

for a system with 2 disks A and B, where it fails if any disk fails:
P(system fails) = P(A or B fails) = $2p$, mean time to failure = $\frac{1}{2p} = \frac{1}{2} MTTF$

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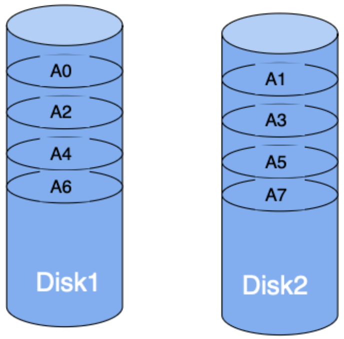
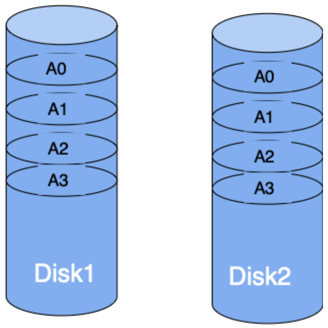
Redundant Array of Independent Disks – different ways to combine multiple disks as a unit for fault tolerance or performance improvement, or both of a database system

RAID 0: Block level Striping
RAID 2: Bit level Striping

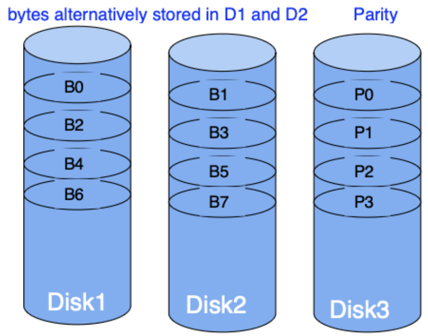


RAID 0 shown here

RAID 1: Mirroring



RAID 3: Parity + Byte level Striping
RAID 4: Parity + Block level Striping



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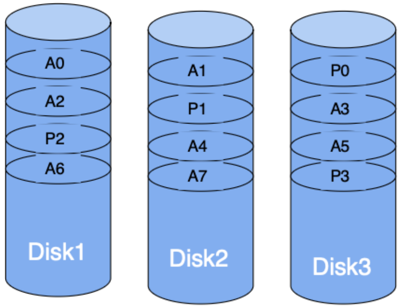
RAID 5: Parity Striping + Block level Striping
RAID 6: 2 parity blocks used (total of 5 disks, can recover from any 2 disk failures)



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Let p = P(one disk failure), and

MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

- File stored across two disks, so if any disk fails, the full file cannot be accessed
- i.e. the system fails if any one of the disks fails.
- P(disk A or disk B fails) = p + p = 2p.
- Mean time to failure = $\frac{1}{2p} = \frac{1}{2} \cdot \frac{1}{p} = \frac{1}{2} \cdot MTTF$

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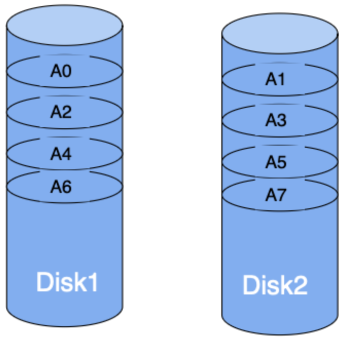
RAID 2: Bit level Striping



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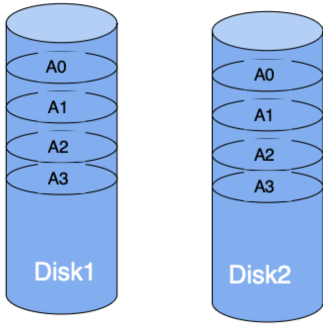
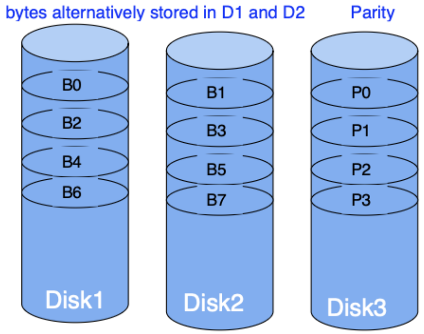


RAID 3: Parity + Byte level Striping
RAID 4: Parity + Block level Striping



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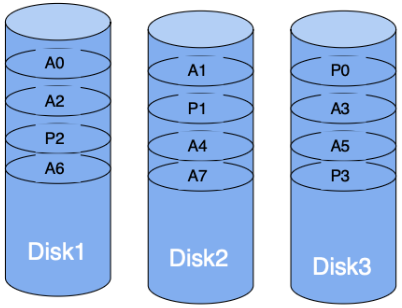
RAID 6: 2 parity blocks used (total of 5 disks, can recover from any 2 disk failures)



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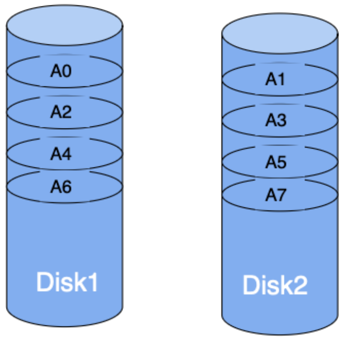
Let p = P(one disk failure), and
MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

- File duplicated, so if one disk fail, the file can still be accessed from another disk
- The system fails only if both disks fail at the same time
- P(disk A and disk B fail) = $p \times p = p^2$.
- Mean time to failure = $\frac{1}{p^2} = \left(\frac{1}{p}\right)^2 = MTTF^2$

RAID 0: Block level Striping
RAID 2: Bit level Striping



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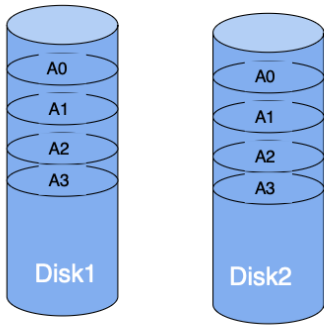

RAID 3: Parity + Byte level Striping
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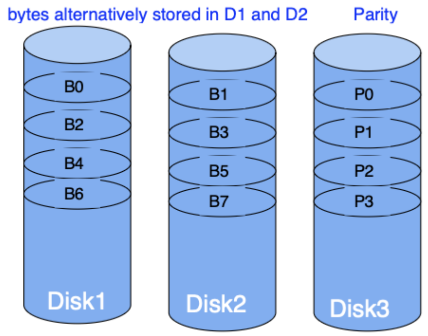

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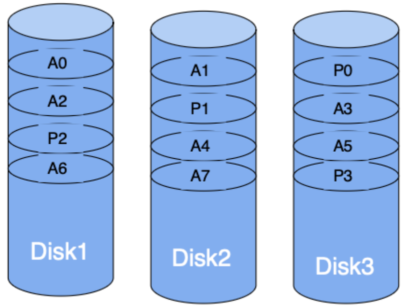

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Let p = P(one disk failure), and

MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

- The system fails only if all three disks fail at the same time
- P(disk A and disk B and disk C fail) = $p \times p \times p = p^3$.
- Mean time to failure = $\frac{1}{p^3} = \left(\frac{1}{p}\right)^3 = MTTF^3$

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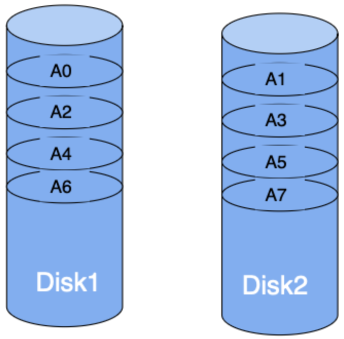
RAID 2: Bit level Striping



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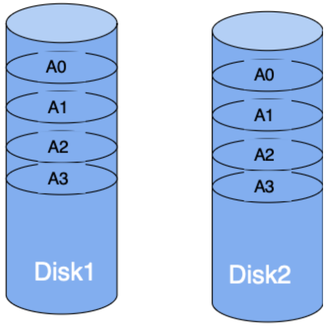

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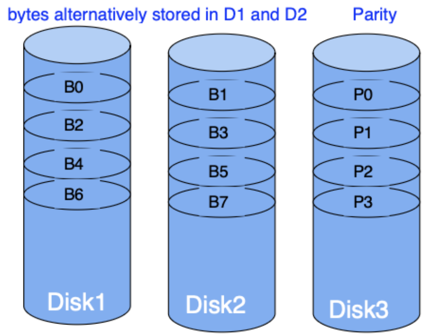

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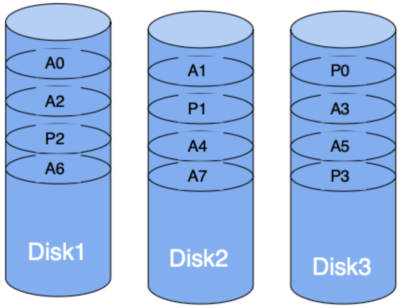

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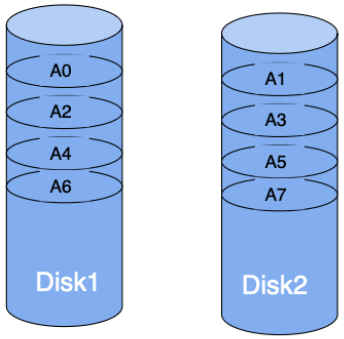
Let p = P(one disk failure), and

MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

- All three systems have a parity, such that if any disk failed, the other two disks can be used to recover it
- The system fails if 2/3 disks fail at the same time
- P(any combinations of 2 disks fail) = $\binom{3}{2} \times p \times p = 3p^2$.
- Mean time to failure = $\frac{1}{3p^2} = \frac{1}{3}\left(\frac{1}{p}\right)^2 = \frac{1}{3} MTTF^2$

RAID 0: Block level Striping
RAID 2: Bit level Striping



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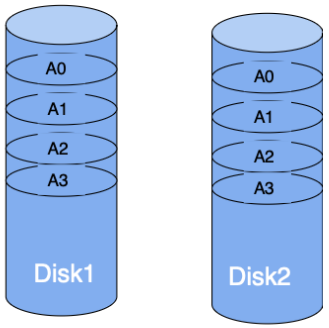

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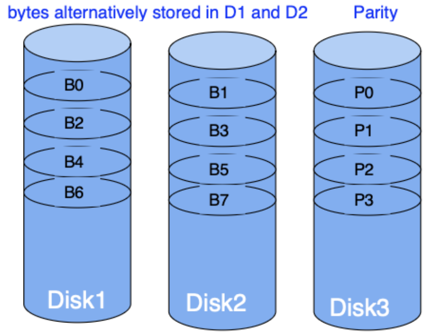

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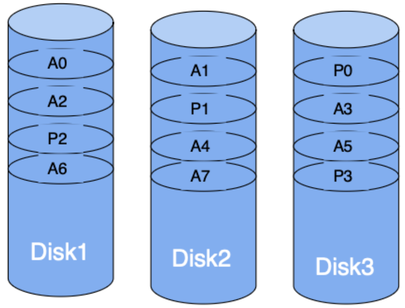

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Let p = P(one disk failure), and

MTTF = $\frac{1}{p}$ = mean time to failure of one individual disk

- The system fails if 3/5 disks fail at the same time
- P(any combinations of 3 disks fail) = $\binom{5}{3} \times p \times p \times p = 10p^3$.
- Mean time to failure = $\frac{1}{10p^3} = \frac{1}{10}\left(\frac{1}{p}\right)^3 = \frac{1}{10} MTTF^3$

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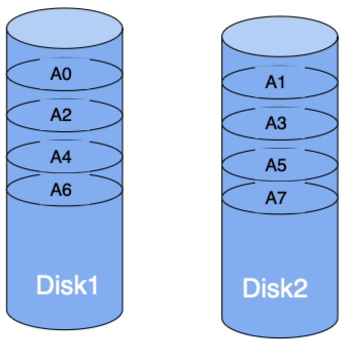

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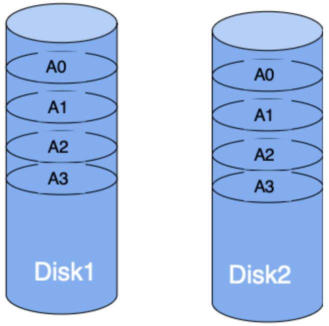

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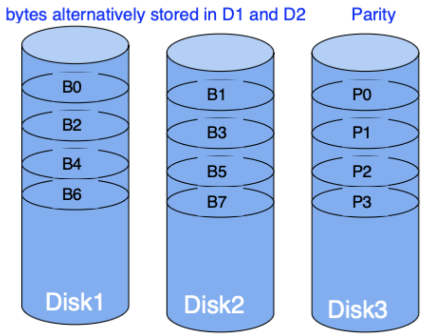

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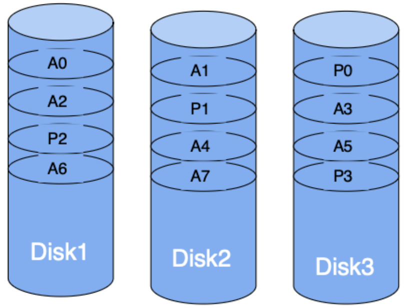
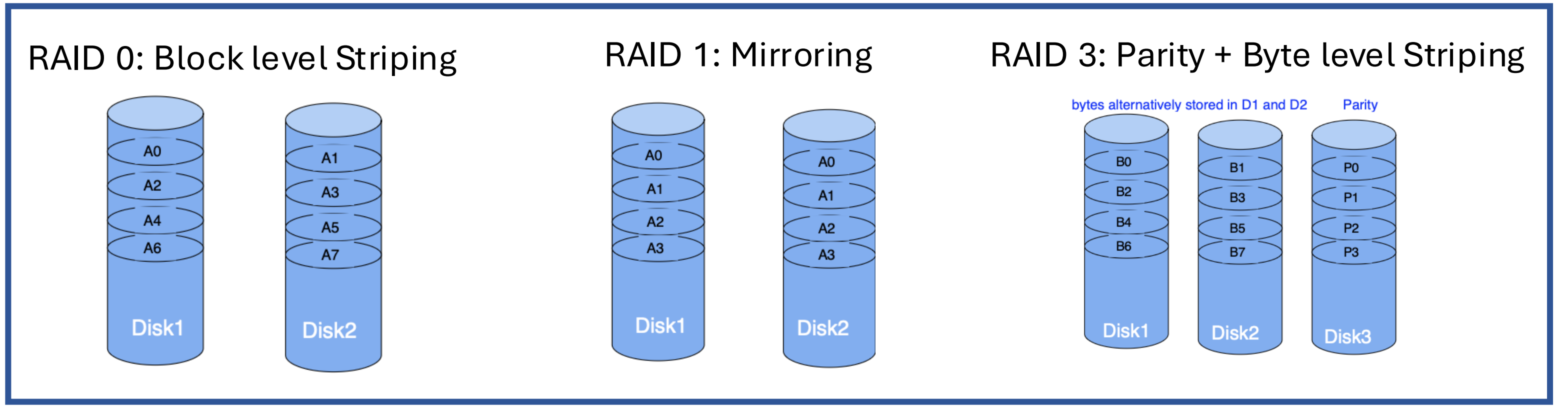


RAID 5 shown here

Q2 RAID and Disk Space Utilization

2. Which of the following RAID configurations that we saw in class has the lowest disk space utilization? Your answer needs to have explanations with calculations for each case.

- (1) RAID 0 with 2 disks
- (2) RAID 1 with 2 disks
- (3) RAID 3 with 3 disks



Where does this lack of utilization of space go, i.e., where we can use such a configuration as it has some benefits gained due to the loss of space utilization?

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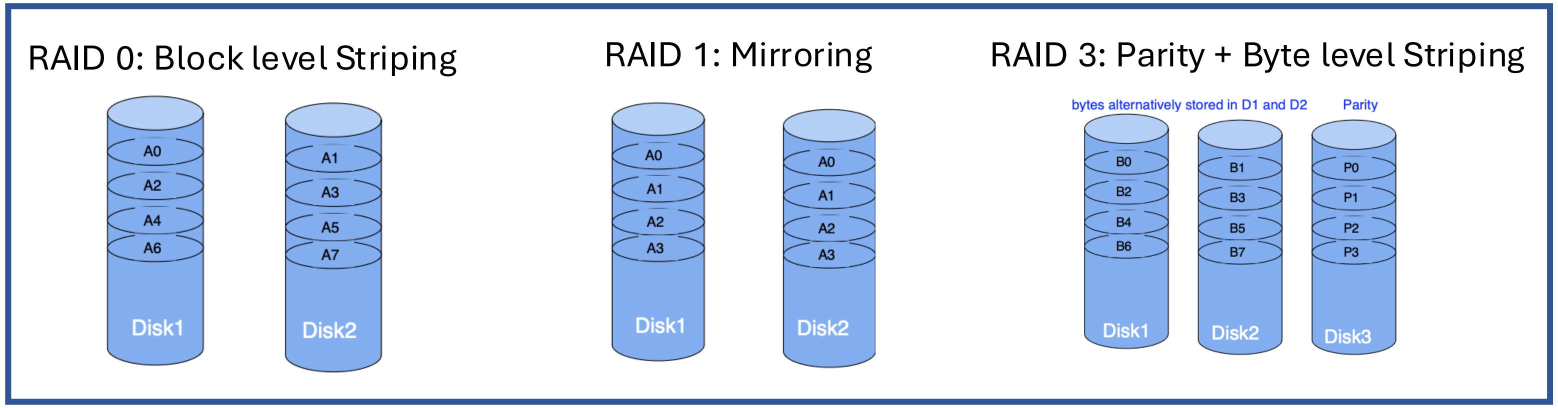

Where does this lack of utilization of space go, i.e., where we can use such a configuration as it has some benefits gained due to the loss of space utilization?

1. Space utilization = 100% because the two disks store contiguous blocks of a file in RAID 0
2. Space utilization = 50% because RAID 1 uses mirroring
3. Space utilization is (3-1)/3=66.7% because RAID 3 uses one disk for storing parity data.

Case 2 has the **lowest disk space utilization**, but has **increased MTTF** as the system operates even when a disk fails. This is good for cases where a disk can fail easily.

Q3 ARIES

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

Analysis phase:
- Scan forward through the log since the latest checkpoint and update DPT and XACT table.
- This recreates the transaction table and the dirty page table at the time of crash.

Redo phase:
- Scan forward through the log starting at the minimum recLSN in DPT, and redo actions that are not yet flushed to disk.
- This makes sure contents of the pages in the buffer pool become exactly what they were at the time of the crash happened.

Undo phase:
- Undo transactions that remain in transaction table after analysis phase with status ≠ committed.
- This cleans up the changes made by all the unfinished transactions

Q3 ARIES - Analysis

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40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
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Scan forward through the log starting at LSN 20.
LSN 25: Initialize XACT table with T1 (LastLSN 10) and T2 (LastLSN 15). Initialize DPT to P1 (RecLSN 10) and P3

DPT:	**XACT:**
P1, recLSN=10	**T1, lastLSN=10, status=running**
P3, recLSN=15	**T2, lastLSN=15, status=running**

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45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

LSN 30: Add (T1, LSN 30) to XACT table. Add (P2, LSN 30) to DPT.

DPT:	XACT:
P1, recLSN=10	T1, **lastLSN=30**, status=running
P3, recLSN=15	T2, lastLSN=15, status=running
P2, recLSN=30	

Q3 ARIES - Analysis

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45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

LSN 35: Change T1 status to "Commit" in XACT table. Set LastLSN=35 for T1 in XACT table.

DPT:
P1, recLSN=10
P3, recLSN=15
P2, recLSN=30

XACT:
T1, **lastLSN=35**, **status=committed**
T2, lastLSN=15, status=running

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45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

LSN 40: Set LastLSN=40 for T2 in XACT table.

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, **lastLSN=40**, status=running
P2, recLSN=30	

Q3 ARIES - Analysis

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35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

LSN 45: Change T2 status to "Abort" in XACT table. Set LastLSN=45 for T2 in XACT table.

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, **lastLSN=45**, **status=aborted**
P2, recLSN=30	

Q3 ARIES - Analysis

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35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

LSN 50: Set LastLSN=50 for T2 in XACT table.

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, **lastLSN=50**, status=aborted
P2, recLSN=30	

Q3 ARIES - Redo

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45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, **recLSN=10**	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, lastLSN=50, status=aborted
P2, recLSN=30	

Scan forward through the log starting at LSN 10 (the smallest recLSN in DPT).

LSN 10:

- Read page P1, check PageLSN stored in the page.
- If PageLSN<10, redo LSN 10 (set value to ZZZ) and set the page's PageLSN=10.

Q3 ARIES - Redo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, lastLSN=50, status=aborted
P2, recLSN=30	

LSN 15:

- Read page P3, check PageLSN stored in the page.
- If PageLSN<15, redo LSN 15 (set value to VVV) and set the page's PageLSN=15.

Q3 ARIES - Redo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, lastLSN=50, status=aborted
P2, recLSN=30	

LSN 30:

- Read page P2, check PageLSN stored in the page.
- If PageLSN<30, redo LSN 30 (set value to XXX) and set the page's PageLSN=30.

Q3 ARIES - Redo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, lastLSN=50, status=aborted
P2, recLSN=30	

LSN 40:

- Read page P1 if it has been flushed, check PageLSN stored in the page.
- If PageLSN < 40, redo LSN 40 (set value to TTT) and set the page's PageLSN=40.

Q3 ARIES - Redo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	T2, lastLSN=50, status=aborted
P2, recLSN=30	

LSN 50:

- Read page P1 if it has been flushed, check PageLSN stored in the page.
- If PageLSN < 50, redo LSN 50 (set value to ZZZ) and set the page's PageLSN=50.

Q3 ARIES - Undo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	**T2, lastLSN=50, status=aborted**
P2, recLSN=30	

T2 (status ≠ commit) must be undone. Put its lastLSN 50 in ToUndo.

LSN 50: Put LSN 15 in ToUndo

Q3 ARIES - Undo

3. Aries Example: After a crash, we find the following log. What will be the analysis, redo, and undo phases?

10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15 T2: UPDATE P3 (OLD: UUU NEW: VVV)
20 BEGIN CHECKPOINT
25 END CHECKPOINT (XACT TABLE=[[T1,10],[T2,15]]; DPT=[[P1,10],[P3,15]])
30 T1: UPDATE P2 (OLD: WWW NEW: XXX)
35 T1: COMMIT
40 T2: UPDATE P1 (OLD: ZZZ NEW: TTT)
45 T2: ABORT
50 T2: CLR P1(ZZZ), undonextLSN=15

DPT:	XACT:
P1, recLSN=10	T1, lastLSN=35, status=committed
P3, recLSN=15	**T2, lastLSN=50, status=aborted**
P2, recLSN=30	

LSN 15:

- Undo LSN 15 - write a CLR for P3 with “set P3=UUU” and undonextLSN=NULL.
- Write UUU into P3.